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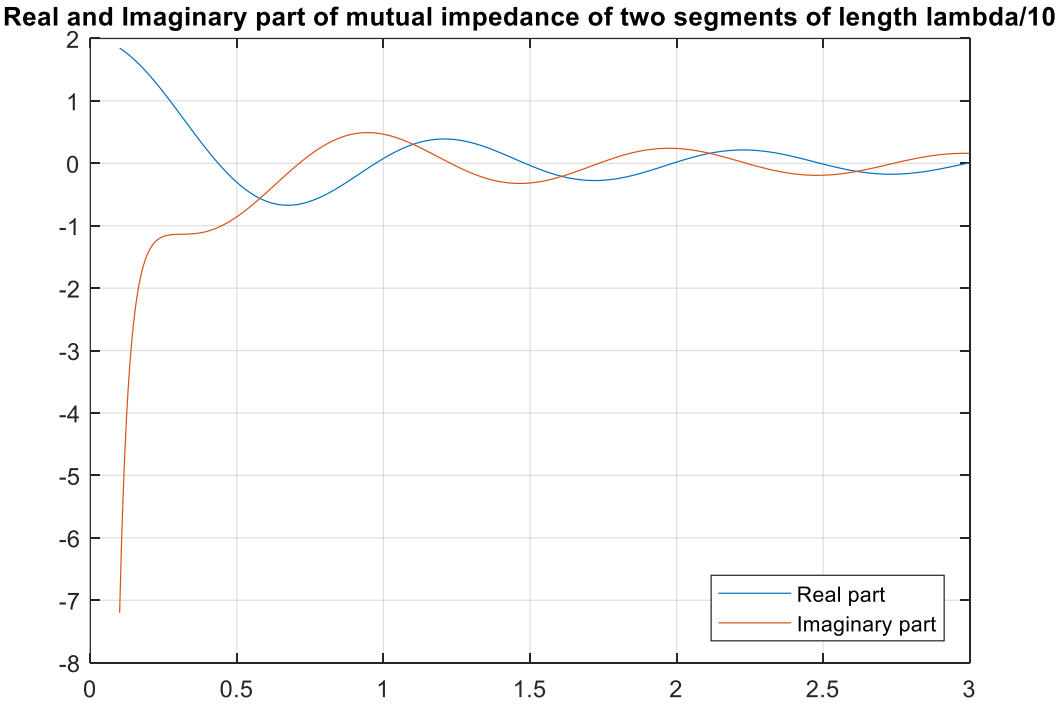
Laboratory 2 Report WLA

1. **Plot the real and imaginary part of the mutual impedance of two segments of length $\lambda/10$ as function of the distance between their centers from 0 to 3λ.**

To do this, we need the following code:

```
%% Exercici 1
% Plot the real and imaginary part of the mutual impedance of two segments of
length
% λ/10 as function of the distance between their centers from 0 to 3 λ.
separation=linspace(0.1,3,500);
l=0.1;
Z=zeros(1,length(separation));
for n1=1:length(separation)
    Z(n1)=z12_dip(l,l,separation(n1),0);
end
figure (1)
plot(separation, real(Z), separation, imag(Z));
title("Real and Imaginary part of mutual impedance of two segments of length
lambda/10")
legend("Real part", "Imaginary part")
grid on;
```

After running the previous code we obtain the following plot:



2. **Write the script.**

We have created a function named “current_distribution” such as it computes the current distribution using the MoM:

```
function [I, zz, II] = current_distribution(lambda_factor,wire_radius,
number_segments)
%This function calculates the current distribution of a dipole throuth the
%MoM
%Inputs:
%   lambda_factor: length of the dipole in relation to the wavelenght
%   wire_radius: wire radius in wavelenght
```

```
%   number_segments: Number of segments. Must be an odd number so we sample
the center.
%Outputs:
%   I: intensity
%   zz: coordinates of the measumremnt points
%   II: current distribution
    H=lambda_factor; %lenght of the dipole arm in relation to the wavelenght
    a=wire_radius; % wire radius in wavelenght
    N=number_segments; %Number of segments. Must be an odd number so we
sample the center.

    l=(2*H)/((N+1)/2);  %lenght of the segment
    z=[-(N-1)/2:(N-1)/2]'*l/2;  %Cordinate of the center of each element.

    Z=zeros(N,N);
    V=zeros(N,1);
    V((N+1)/2)=1;   %We use a feed voltage of 1V

    for n1=1:N
        Z(n1,n1)=z12_dip(l,l,a,0);
        for n2=n1+1:N
            Z(n1, n2)=z12_dip(l,l,a,(z(n2)-z(n1)));
            Z(n2,n1)=Z(n1,n2);
        end
    end

    I=inv(Z)*V;

    zz=[z(1)-l/2;z;z(N)+l/2];
    II=[0;I;0];
end
```

3. Check your results for a half-wave dipole.

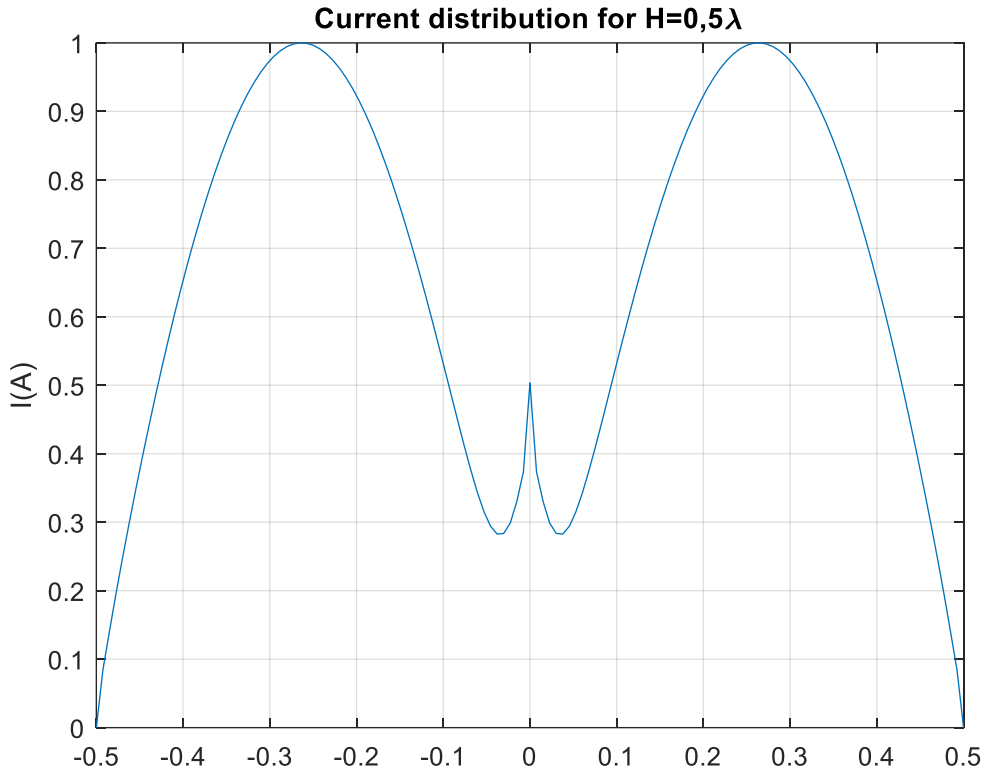
Using the previous function we compute the results for a half-wave dipole with the following script:

```
%% Exercici 2 & 3
H=0.5;
a=0.001;
N=131;
[I, zz, II]=current_distribution(H,a,N);
figure(2);
plot(zz, abs(II)/max(max(abs(II))));
title("Current distribution for H=0,5\lambda")
ylabel("I(A)")
grid on;
Zin=1/I((N+1)/2)
```

```
Zin =

   7.1449e+02 - 1.0457e+03i
```

We obtain a Zin of $7.1449e+02 - 1.0457e+03i$ and the following plot:



4. **Compare the current distribution for a dipole H=0.1λ, 0.25λ and 5/8*λ with the transmission line model.**

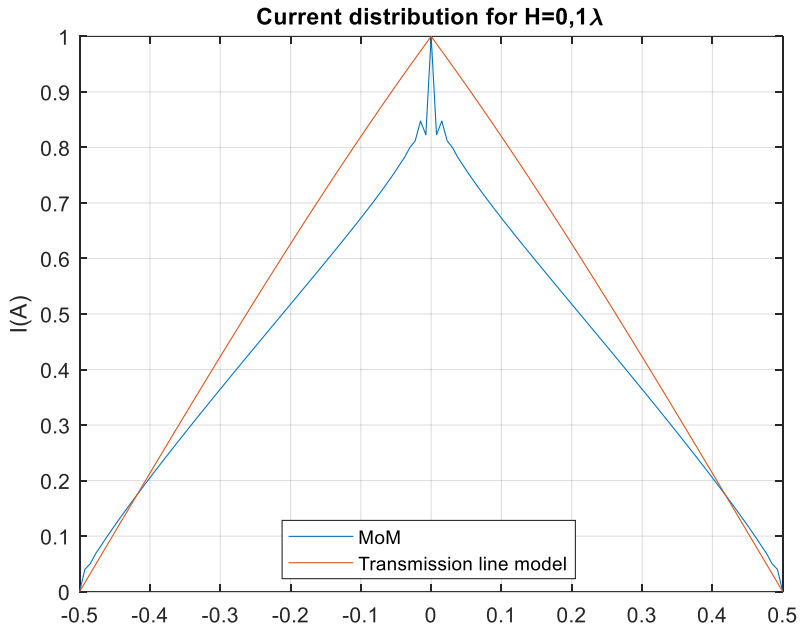
To be able to compare the current distribution with the MoM and the transmission line model we use the following script:

```
%% Exercici 4
%Case 1: 0.1*lambda
H_1=0.1;
[I_1, zz_1, II_1]=current_distribution(H_1,a,N);
I_TL_1 = sin(2*pi*(H_1-abs(zz_1)));
figure (3);
plot(zz, abs(II_1)/max(max(abs(II_1))),zz,
abs(I_TL_1)/max(max(abs(I_TL_1))));
title("Current distribution for H=0,1\lambda")
ylabel("I(A)")
grid on;
legend("MoM", "Transmission line model")
Zin_1=1/I_1((N+1)/2)
%Case 2:0.25*lambda
H_2=0.25;
[I_2, zz_2, II_2]=current_distribution(H_2,a,N);
I_TL_2 = sin(2*pi*(H_2-abs(zz_2)));
figure (4);
plot(zz_2, abs(II_2)/max(max(abs(II_2))), zz_2,
abs(I_TL_2)/max(max(abs(I_TL_2))));
title("Current distribution for H=0,25\lambda")
ylabel("I(A)")
grid on;
legend("MoM", "Transmission line model")
Zin_2=1/I_2((N+1)/2)
%Case 3: (5/8)*lambda
H_3=5/8;
```

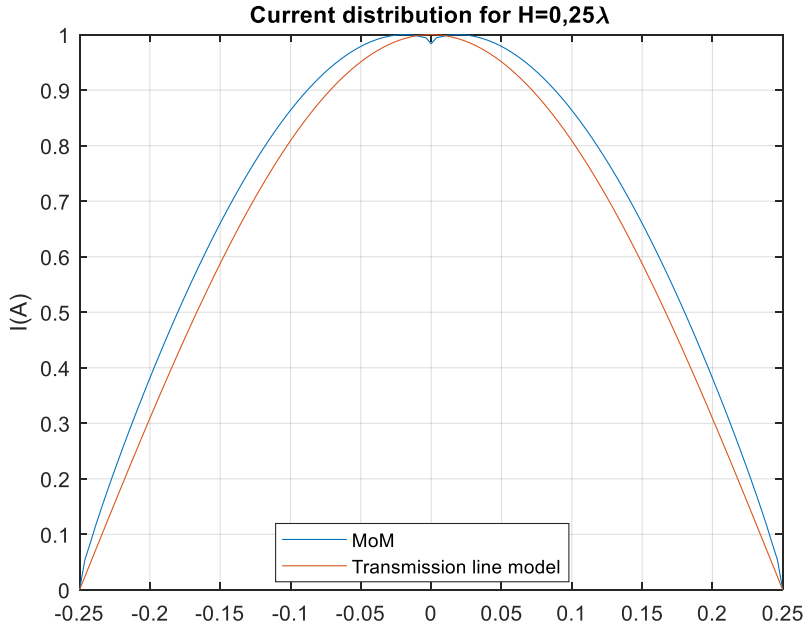
```
[I_3, zz_3, II_3]=current_distribution(H_3,a,N);
I_TL_3 = sin(2*pi*(H_3-abs(zz_3)));
figure (5);
plot(zz_3, abs(II_3)/max(max(abs(II_3))), zz_3,
abs(I_TL_3)/max(max(abs(I_TL_3))));
title("Current distribution for H=(5/8)\lambda")
ylabel("I(A)")
grid on;
legend("MoM", "Transmission line model")
Zin_3=1/I_3((N+1)/2)
```

Running the scripts we can see the following plots:

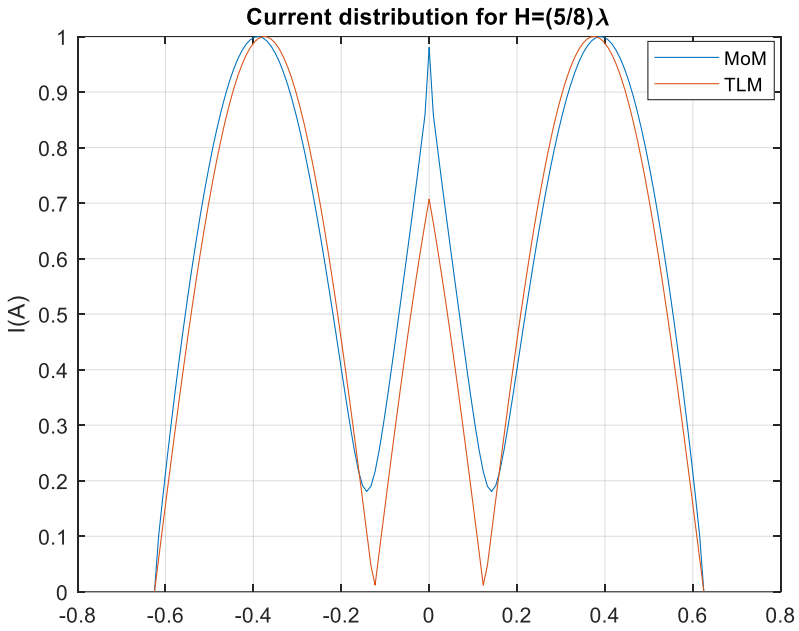
- Case 1: 0.1λ:
  In this case we can observe how the transmission line model has less detail than the MoM version



- Case 2: 0.25λ



- Case 3: (5/8)λ



In all of the cases we can observe that the plot of the MoM and the TLM are very similar.
The impedance at the entrance for the different cases are:

```
Zin_1 =

   6.0535e+00 - 5.0304e+02i


Zin_2 =

  86.9767 +47.0863i


Zin_3 =

   1.0324e+02 - 4.1576e+02i
```

5. Use your scripts from the previous lab to obtain the radiation pattern and directivity.

To plot the radiation pattern and the maximum directivity for different H we use the following script:

```
%% Exercici 5
aux = [0.1, 0.25, 0.5, 5/8];
for v = aux
    H=v;

    %MoM
    [I,zz,II]=current_distribution(H,a,N);
    II_norm = II/norm(II);
    Eth = zeros(size(th));
    Eph = zeros(size(ph));
    for f=1:length(II)
        Eth = Eth+sin(th).*II(f,:).*exp(1i*2*pi*cos(th).*zz(f,:));
    end
    P = abs(Eth).^2+abs(Eph).^2;
    t = P/max(max(P));
```

```
    D = (4*pi)/(sum(sum(t.*sin(th)))*inc_th*inc_ph);

    %Transmission line
    Eth_tl = zeros(size(th));
    Eph_tl = zeros(size(ph));
    II_tl = sin(2*pi*(H-abs(zz)));
    II_tl_nomr=II_tl/norm(II_tl);
    for f_tl=1:length(II_tl)
        Eth_tl =
Eth_tl+sin(th).*II_tl(f_tl,:).*exp(j*2*pi*cos(th).*zz(f_tl,:));
    end
    P_tl = abs(Eth_tl).^2+abs(Eph_tl).^2;
    t_tl = P_tl/max(max(P_tl));
    D_tl = (4*pi)/(sum(sum(t_tl.*sin(th)))*inc_th*inc_ph);

    disp("The directivity of H=");
    disp(H);
    disp(" using the MoM model is D=");
    disp(D);
    disp("and using the TL model is D_tl=");
    disp(D_tl);

    %Plots
    figure();
    [ang,E] = extrae_corte(t,'XY');
    polar_xy(ang,E,1,0,5,12,'b');
    title("H-Plane H="+v+"\lambda");
    hold on
    [ang_tl,E_tl] = extrae_corte(t_tl,'XY');
    polar_xy(ang_tl,E_tl,1,0,5,12,'r');
    legend('MoM','TL');
    hold off

    figure();
    [ang,E] = extrae_corte(t,'ZX');
    polar_zx(ang,E,1,0,5,12,'b');
    title("E-Plane H="+v+"\lambda");
    hold on
    [ang_tl,E_tl] = extrae_corte(t_tl,'ZX');
    polar_zx(ang_tl,E_tl,1,0,5,12,'r');
    legend('MoM','TL');
    hold off
end
```

Using this script, we obtain the following maximum values of directivity:

```
The directivity of H=
    0.1000

 using the MoM model is D=
    1.5211

and using the TL model is D_tl=
    1.5201

The directivity of H=
    0.2500

 using the MoM model is D=
    1.6514

and using the TL model is D_tl=
    1.6404

The directivity of H=
    0.5000

 using the MoM model is D=
    2.4809

and using the TL model is D_tl=
    2.4095

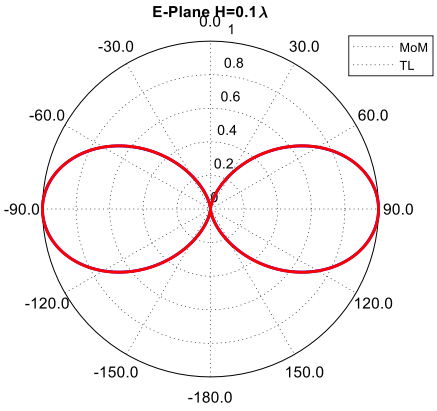
The directivity of H=
    0.6250

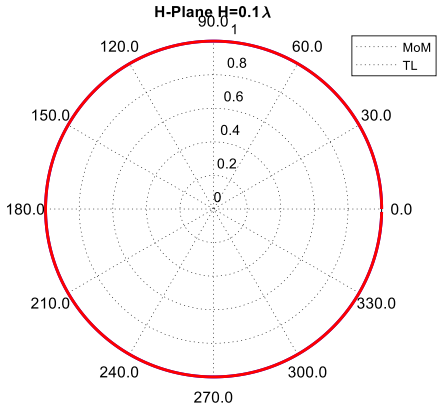
 using the MoM model is D=
    3.1592

and using the TL model is D_tl=
    3.2795
```

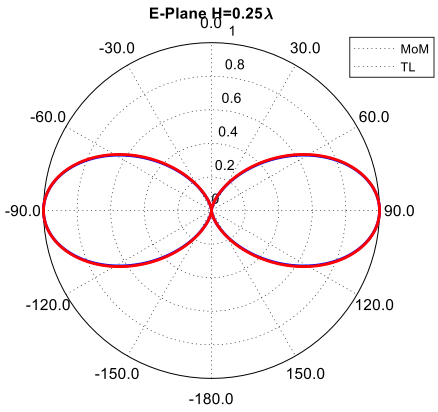
We also obtain the different plots of radiation for the E and H fields for the different H introduced:

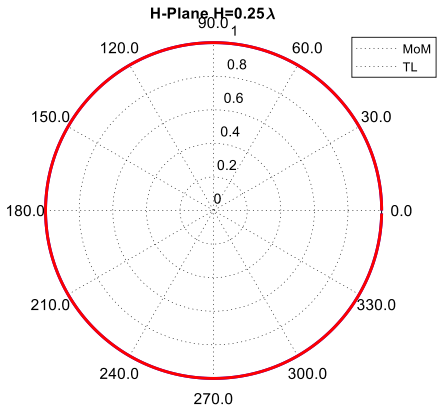
- For 0.1λ:



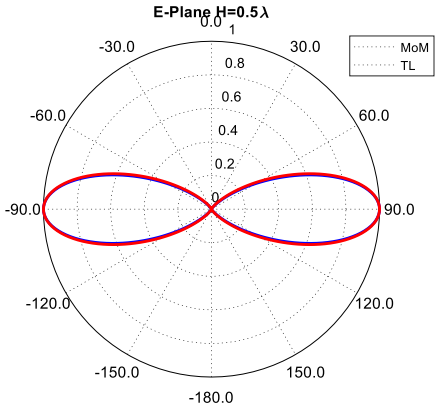


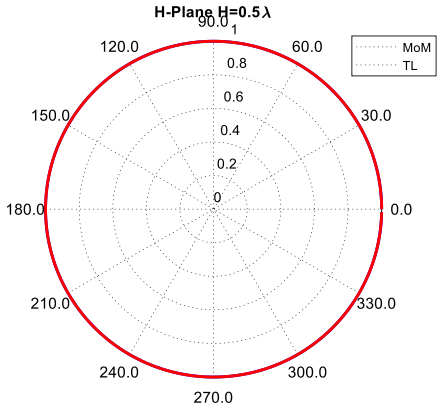
- For 0.25λ:



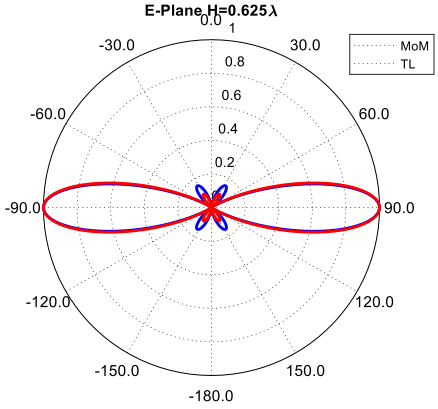


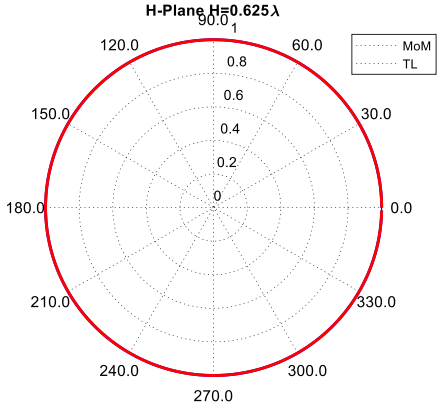
- For 0.5λ:





- For (5/8)λ:





6. **Obtain a plot for the directivity as function of 2H/λ from 0.1 to 1.5 using the transmission line model and the MoM.**

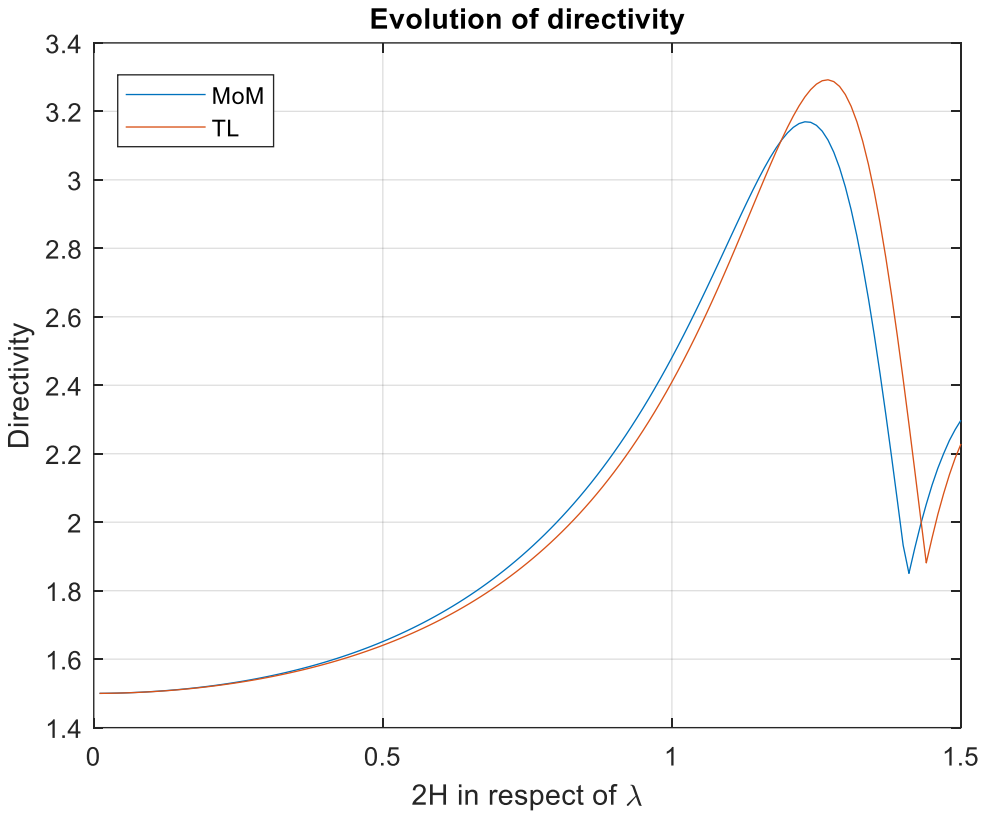
To obtain the plot directivity of as a function of 2H we need the following script:

```
%% Exercici 6
H_6=(0.01:0.01:1.5)/2;
N_6=31;
Z = zeros(N,N);
V = zeros(N,1);
V((N+1)/2) = 1;
D=zeros(size(H_6));
D_tl=zeros(size(H_6));
```

```
for h=1:length(H_6)
    %MoM
    l = (2*H_6(:,h))/((N_6+1)/2);
    z = [-(N_6-1)/2:(N_6-1)/2]'*l/2;
    Eth = zeros(size(th));
    Eph = zeros(size(ph));
    for n1 = 1:N_6
        Z(n1,n1) = z12_dip(l,l,a,0);
        for n2 = n1+1:N_6
            Z(n1,n2) = z12_dip(l,l,a,(z(n2)-z(n1)));
            Z(n2,n1) = Z(n1,n2);
        end
    end
    I = inv(Z)*V;
    zz = [z(1)-l/2;z;z(N_6)+l/2];
    II = [0;I;0];
    II_norm = II/norm(II);
    for f=1:length(II)
       Eth = Eth+sin(th).*II(f,:).*exp(j*2*pi*cos(th).*zz(f,:));
    end
    P = abs(Eth).^2+abs(Eph).^2;
    t = P/max(max(P));
    D(:,h) = (4*pi)/(sum(sum(t.*sin(th)))*inc_th*inc_ph);

    %Transmission line
    Eth_tl = zeros(size(th));
    Eph_tl = zeros(size(ph));
    II_tl = sin(2*pi*(H_6(:,h)-abs(zz)));
    II_tl_nomr=II_tl/norm(II_tl);
    for f_tl=1:length(II_tl)
        Eth_tl =
Eth_tl+sin(th).*II_tl(f_tl,:).*exp(j*2*pi*cos(th).*zz(f_tl,:));
    end
    P_tl = abs(Eth_tl).^2+abs(Eph_tl).^2;
    t_tl = P_tl/max(max(P_tl));
    D_tl(:,h) = (4*pi)/(sum(sum(t_tl.*sin(th)))*inc_th*inc_ph);
end
figure();
plot(2*H_6,D,2*H_6,D_tl);
title("Evolution of directivity");
xlabel('2H in respect of \lambda');
ylabel('Directivity');
legend('MoM', 'TL');
grid on;
```

This scrip produces the following plot:


Evolution of directivity
MoM
TL
Directivity
2H in respect of λ
3.4
3.2
3
2.8
2.6
2.4
2.2
2
1.8
1.6
1.4
0
0.5
1
1.5